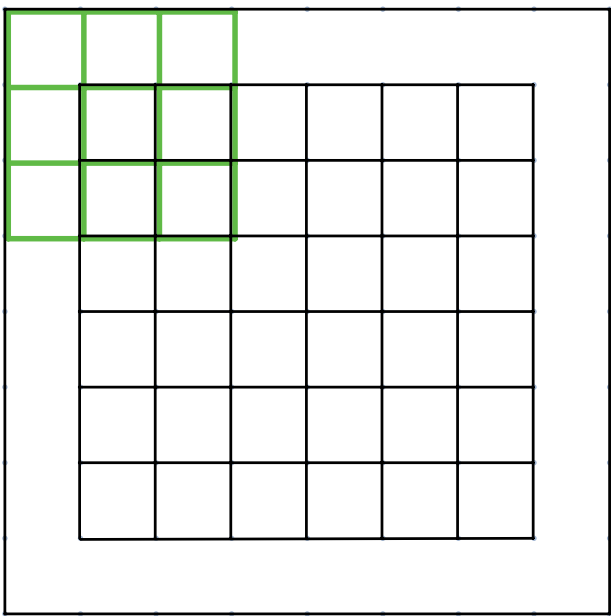
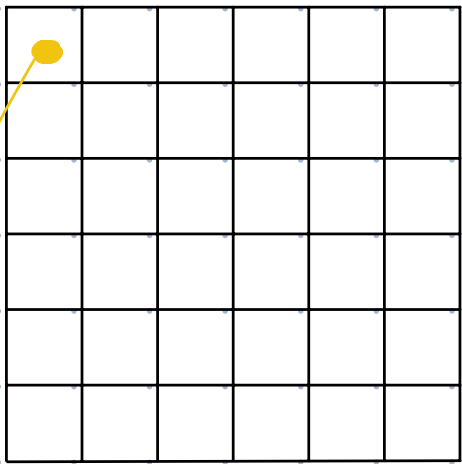




0,0



$$\frac{3-1}{2} = 1$$

$$(f_i, f_j) = (0, 0)$$

$$(row_hat, col_hat) = (\quad)$$

0,0

0,0	0,1	0,2			
1,0	1,1	1,2			
2,0	2,1	2,2			

0,0		

$$f_i = 0 \rightarrow 2$$

$$f_j = 0 \rightarrow 2$$

$$row' = 0 - 1 + 0 + 1$$

$$col' =$$

Formula:

$$row' = row - radius + f_i + pad_h$$

$$col' = col - radius + f_j + pad_w$$

z_1	z_2
z_4	z_5

$\times N_1$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_1} * x_1 + \frac{\partial L}{\partial z_2} * x_2 + \frac{\partial L}{\partial z_3} * x_5 + \frac{\partial L}{\partial z_4} * x_6$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial z_1} * x_2 + \frac{\partial L}{\partial z_2} * x_3 + \frac{\partial L}{\partial z_3} * x_6 + \frac{\partial L}{\partial z_4} * x_7$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial z_1} * x_3 + \frac{\partial L}{\partial z_2} * x_4 + \frac{\partial L}{\partial z_3} * x_7 + \frac{\partial L}{\partial z_4} * x_8$$

$$\frac{\partial L}{\partial w_4} = \frac{\partial L}{\partial z_1} * x_5 + \frac{\partial L}{\partial z_2} * x_6 + \frac{\partial L}{\partial z_3} * x_9 + \frac{\partial L}{\partial z_4} * x_{10}$$

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial z_1} * x_6 + \frac{\partial L}{\partial z_2} * x_7 + \frac{\partial L}{\partial z_3} * x_{10} + \frac{\partial L}{\partial z_4} * x_{11}$$

$$\frac{\partial L}{\partial w_6} = \frac{\partial L}{\partial z_1} * x_7 + \frac{\partial L}{\partial z_2} * x_8 + \frac{\partial L}{\partial z_3} * x_{11} + \frac{\partial L}{\partial z_4} * x_{12}$$

$$\frac{\partial L}{\partial w_7} = \frac{\partial L}{\partial z_1} * x_9 + \frac{\partial L}{\partial z_2} * x_{10} + \frac{\partial L}{\partial z_3} * x_{13} + \frac{\partial L}{\partial z_4} * x_{14}$$

$$\frac{\partial L}{\partial w_8} = \frac{\partial L}{\partial z_1} * x_{10} + \frac{\partial L}{\partial z_2} * x_{11} + \frac{\partial L}{\partial z_3} * x_{14} + \frac{\partial L}{\partial z_4} * x_{15}$$

$$\frac{\partial L}{\partial w_9} = \frac{\partial L}{\partial z_1} * x_{11} + \frac{\partial L}{\partial z_2} * x_{12} + \frac{\partial L}{\partial z_3} * x_{15} + \frac{\partial L}{\partial z_4} * x_{16}$$

$\frac{\partial L}{\partial z_1}$	$\frac{\partial L}{\partial z_2}$
$\frac{\partial L}{\partial z_3}$	$\frac{\partial L}{\partial z_4}$

we call this
b-dot

Complete backprop for weight

$$\begin{array}{c} \frac{\partial L}{\partial w_1} \quad \frac{\partial L}{\partial w_2} \quad \frac{\partial L}{\partial w_3} \\ \frac{\partial L}{\partial w_4} \quad \frac{\partial L}{\partial w_5} \quad \frac{\partial L}{\partial w_6} \\ \frac{\partial L}{\partial w_7} \quad \frac{\partial L}{\partial w_8} \quad \frac{\partial L}{\partial w_9} \end{array} = \begin{array}{|c|c|c|} \hline x_1 & x_2 & x_3 \\ \hline x_5 & x_6 & x_7 \\ \hline x_9 & x_{10} & x_{11} \\ \hline \end{array} * \frac{\partial L}{\partial z_1} + \begin{array}{|c|c|c|} \hline x_2 & x_3 & x_4 \\ \hline x_6 & x_7 & x_8 \\ \hline x_{10} & x_{11} & x_{12} \\ \hline \end{array} * \frac{\partial L}{\partial z_2} + \begin{array}{|c|c|c|} \hline x_5 & x_6 & x_7 \\ \hline x_9 & x_{10} & x_{11} \\ \hline x_{13} & x_{14} & x_{15} \\ \hline \end{array} * \frac{\partial L}{\partial z_3} + \begin{array}{|c|c|c|} \hline x_6 & x_7 & x_8 \\ \hline x_{10} & x_{11} & x_{12} \\ \hline x_{14} & x_{15} & x_{16} \\ \hline \end{array} * \frac{\partial L}{\partial z_4}$$

If there is more than one image, then also sum across batch.

Similarly, complete back prop for input

at substitution

$$\begin{aligned} \frac{\partial L}{\partial X_1} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_1} \\ \frac{\partial L}{\partial X_2} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_2} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_2} \\ \frac{\partial L}{\partial X_3} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_3} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_3} \\ \frac{\partial L}{\partial X_4} &= \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_4} \\ \frac{\partial L}{\partial X_5} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_5} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_5} \\ \frac{\partial L}{\partial X_6} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_6} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_6} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_6} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_6} \\ \frac{\partial L}{\partial X_7} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_7} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_7} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_7} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_7} \\ \frac{\partial L}{\partial X_8} &= \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_8} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_8} \\ \frac{\partial L}{\partial X_9} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_9} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_9} \\ \frac{\partial L}{\partial X_{10}} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_{10}} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_{10}} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_{10}} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_{10}} \\ \frac{\partial L}{\partial X_{11}} &= \frac{\partial L}{\partial z_1} \times \frac{\partial z_1}{\partial X_{11}} + \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_{11}} + \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_{11}} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_{11}} \\ \frac{\partial L}{\partial X_{12}} &= \frac{\partial L}{\partial z_2} \times \frac{\partial z_2}{\partial X_{12}} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_{12}} \\ \frac{\partial L}{\partial X_{13}} &= \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_{13}} \\ \frac{\partial L}{\partial X_{14}} &= \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_{14}} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_{14}} \\ \frac{\partial L}{\partial X_{15}} &= \frac{\partial L}{\partial z_3} \times \frac{\partial z_3}{\partial X_{15}} + \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial X_{15}} \\ \frac{\partial L}{\partial z_4} &= \frac{\partial L}{\partial z_4} \times \frac{\partial z_4}{\partial z_4} \end{aligned}$$

$$\frac{\partial X_{16}}{\partial z_4} = \frac{\partial L}{\partial z_4} \times \frac{\partial X_{16}}{\partial L}$$

$$\frac{\partial L}{\partial X_1} = \frac{\partial L}{\partial z_1} \times w_1$$

$$\frac{\partial L}{\partial X_2} = \frac{\partial L}{\partial z_1} \times w_2 + \frac{\partial L}{\partial z_2} \times w_1$$

$$\frac{\partial L}{\partial X_3} = \frac{\partial L}{\partial z_1} \times w_3 + \frac{\partial L}{\partial z_2} \times w_2$$

$$\frac{\partial L}{\partial X_4} = \frac{\partial L}{\partial z_2} \times w_3$$

$$\frac{\partial L}{\partial X_5} = \frac{\partial L}{\partial z_1} \times w_4 + \frac{\partial L}{\partial z_3} \times w_1$$

$$\frac{\partial L}{\partial X_6} = \frac{\partial L}{\partial z_1} \times w_5 + \frac{\partial L}{\partial z_2} \times w_4 + \frac{\partial L}{\partial z_3} \times w_2 + \frac{\partial L}{\partial z_4} \times w_1$$

$$\frac{\partial L}{\partial X_7} = \frac{\partial L}{\partial z_1} \times w_6 + \frac{\partial L}{\partial z_2} \times w_5 + \frac{\partial L}{\partial z_3} \times w_3 + \frac{\partial L}{\partial z_4} \times w_2$$

$$\frac{\partial L}{\partial X_8} = \frac{\partial L}{\partial z_2} \times w_6 + \frac{\partial L}{\partial z_4} \times w_3$$

$$\frac{\partial L}{\partial X_9} = \frac{\partial L}{\partial z_1} \times w_7 + \frac{\partial L}{\partial z_3} \times w_4$$

$$\frac{\partial L}{\partial X_{10}} = \frac{\partial L}{\partial z_1} \times w_8 + \frac{\partial L}{\partial z_2} \times w_7 + \frac{\partial L}{\partial z_3} \times w_5 + \frac{\partial L}{\partial z_4} \times w_4$$

$$\frac{\partial L}{\partial X_{11}} = \frac{\partial L}{\partial z_1} \times w_9 + \frac{\partial L}{\partial z_2} \times w_8 + \frac{\partial L}{\partial z_3} \times w_6 + \frac{\partial L}{\partial z_4} \times w_5$$

$$\frac{\partial L}{\partial X_{12}} = \frac{\partial L}{\partial z_2} \times w_9 + \frac{\partial L}{\partial z_4} \times w_6$$

$$\frac{\partial L}{\partial X_{13}} = \frac{\partial L}{\partial z_3} \times w_7$$

$$\frac{\partial L}{\partial X_{14}} = \frac{\partial L}{\partial z_3} \times w_8 + \frac{\partial L}{\partial z_4} \times w_7$$

$$\frac{\partial L}{\partial X_{15}} = \frac{\partial L}{\partial z_3} \times w_9 + \frac{\partial L}{\partial z_4} \times w_8$$

$$\frac{\partial L}{\partial X_{16}} = \frac{\partial L}{\partial z_4} \times w_9$$

$\frac{\partial L}{\partial z_1}$	$\frac{\partial L}{\partial z_2}$
$\frac{\partial L}{\partial z_3}$	$\frac{\partial L}{\partial z_4}$

$\ast \omega_1$

+

$\frac{\partial L}{\partial z_1}$	$\frac{\partial L}{\partial z_2}$
$\frac{\partial L}{\partial z_3}$	$\frac{\partial L}{\partial z_4}$

$\ast \omega_2$

+

$\frac{\partial L}{\partial z_1}$	$\frac{\partial L}{\partial z_2}$
$\frac{\partial L}{\partial z_3}$	$\frac{\partial L}{\partial z_4}$

$\ast \omega_3$

for $k_y \rightarrow k$

for $k_x \rightarrow k$

Take $w-i$

$$\text{grad_in} = [B, C_{in}, H_{in}, W_{in}]$$

$$\text{filters} = [C_{out}, C_{in}, k, k]$$

$$\text{dout} = [B, C_{out}, H_{out}, W_{out}]$$

x_1	x_2	x_3	x_4
x_5	x_6	x_7	x_8
x_9	x_{10}	x_{11}	x_{12}
x_{13}	x_{14}	x_{15}	x_{16}

d-in

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

kernel

dout

for $k_y \rightarrow K$:

for $k_x \rightarrow k$:

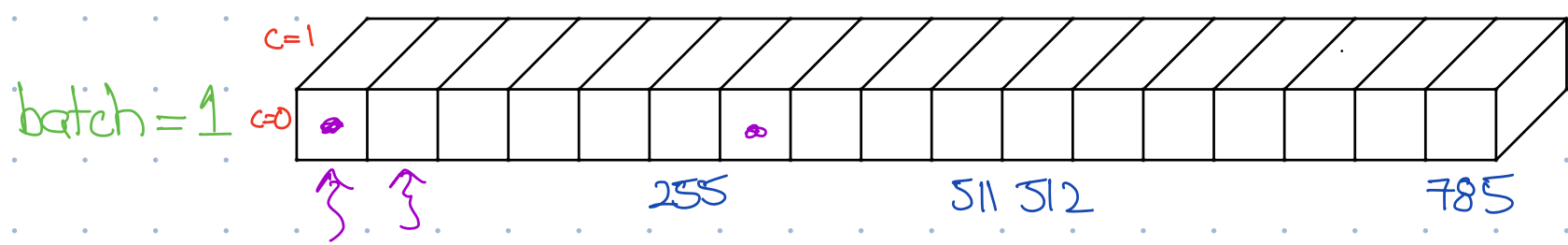
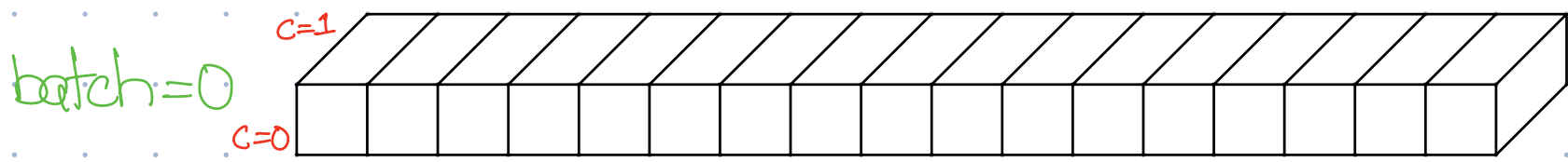
$$\text{dout_x} = \text{din_x} - k_x + \text{pad_w} \quad 0 - 0 + 0$$

$$\text{dout_y} = \text{din_y} - k_y + \text{pad_h} \quad 0 - 0 + 0$$

$$w_i = \text{kernel}[k_x, k_y] \quad \text{kernel}[0, 0]$$

$$d = \text{dout}[\text{dout_y}, \text{dout_x}] \quad \text{dout}[0, 0]$$

$$\text{grad} = w_i \ast d$$



Block_DIM = 256

Num blocks = $\text{ceil}\left(\frac{784}{256 \times 2}\right) = 2$

(2, 4)

batch_idx = $0 / 2 = 0$

channel_idx = $0 \% 2 = 0$

785

512

512 + 256

Fused softmax kernel

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_i e^{z_i}}$$

$$\mathcal{L} = - \sum_{i=1}^C y_i \ln(\sigma(z_i))$$

$$= - \sum_{i=1}^C y_i \ln \left[\frac{e^{z_i}}{\sum_i e^{z_i}} \right]$$

$$= - \sum_{i=1}^C y_i \left[\ln(e^{z_i}) - \ln \left(\sum_i e^{z_i} \right) \right]$$

$$= - \sum_{i=1}^C y_i \left[z_i - \ln \left(\sum_i e^{z_i} \right) \right]$$

As y_i is label index $[1, 2, 0 \dots]$ we can simplify

$$= -z_k + \ln \left(\sum_i e^{z_i} \right)$$

For numerical stability, we do:

$$= -(z_k - M) + \ln \left(\sum_i e^{z_i - M} \right)$$

where M is $\max_C z$

For backward pass:

$$\mathcal{L} = -z_k + \ln \left(\sum_i e^{z_i} \right)$$

Case 01: target logit

$$\frac{dL}{dz_k} = -1 + \frac{e^{z_i}}{\sum_i e^{z_i}}$$

Case 01: not target logit

$$\frac{dL}{z_j \neq k} = 0 + \frac{e^{z_i}}{\sum_i e^{z_i}}$$

$$y = [0, 2, 3]$$

logits

grad-z [BS, classes]

1
3

$$\text{grad-z}[i] = \frac{\text{logits}[i]}{S}$$

if $i = y[bs]$

$$-1 + \frac{\text{logits}[i]}{S}$$

Autograd

data;
grad;
prev

parents: {input, filter}

backward_func(Node* self, Tensor t)

backward_func(Node* self, Tensor t) {